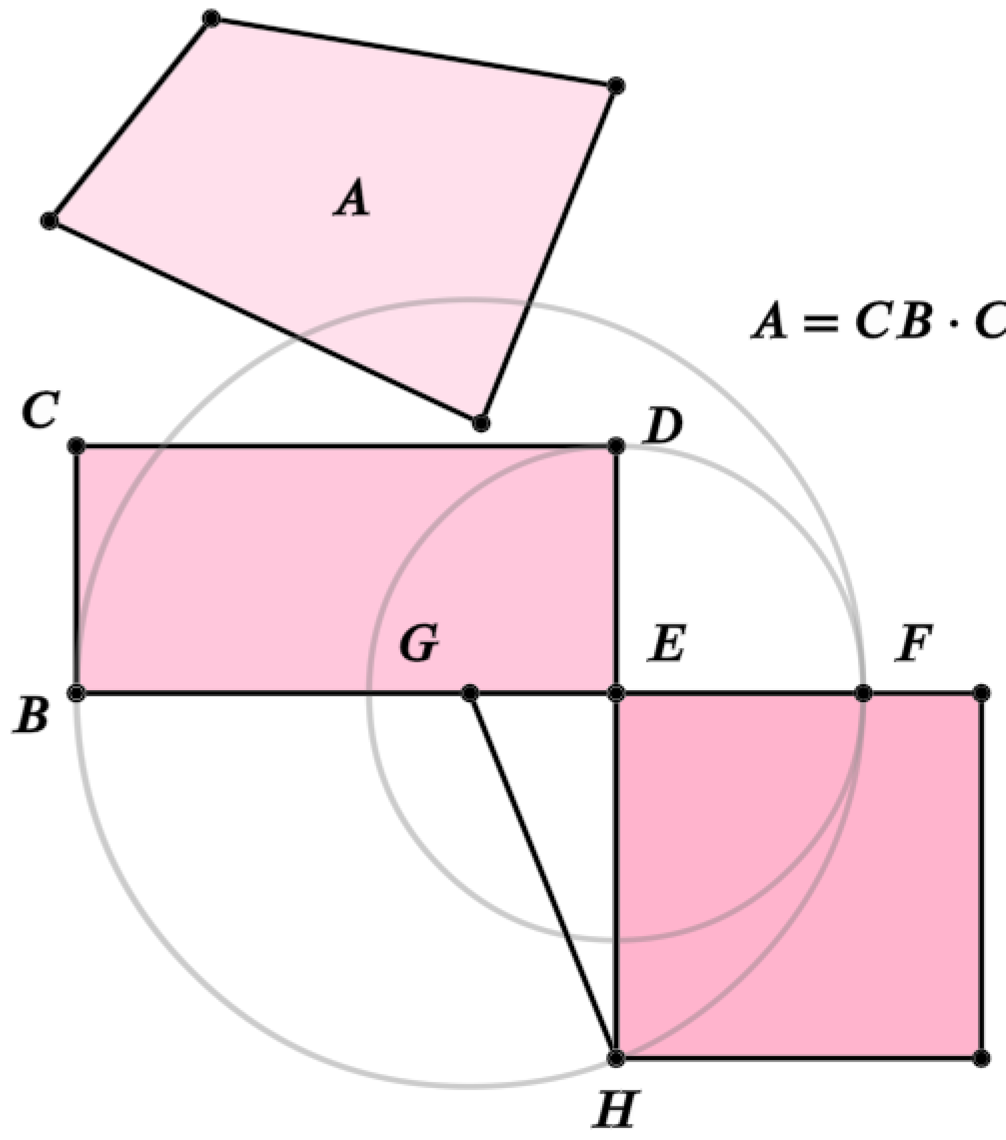
Euclid's Elements

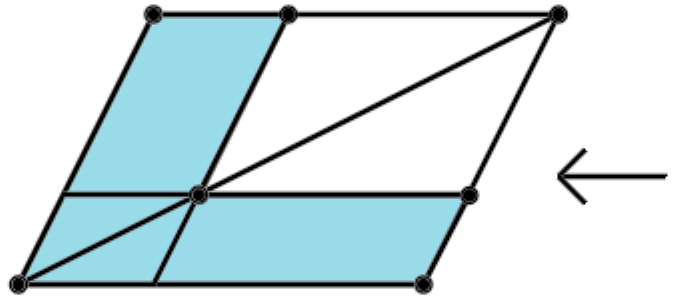
Book II



$$A = CB \cdot CD = (EH)^2$$

It is a remarkable fact in the history of geometry, that the Elements of Euclid, written two thousand years ago, are still regarded by many as the best introduction to the mathematical sciences.

- Florian Cajori,
A History of Mathematics (1893)



Definitions:

Any rectangular parallelogram is said to be contained by the two straight lines containing the right angle.

And in any parallelogrammic area let any one whatever of the parallelograms about its diameter with the two complements be called a gnomon.

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Proposition 8 of Book 2

If a straight line be cut at random, four times the rectangle contained by the whole and one of the segments together with the square on the remaining segment is equal to the square described on the whole and aforesaid segment as on one straight line.

In other words

Let AB be a straight line, arbitrarily cut at point C

$AB = AC + CB$

$AB = x + y$

Proposition 8 of Book 2

If a straight line be cut at random, four times the rectangle contained by the whole and one of the segments together with the square on the remaining segment is equal to the square described on the whole and aforesaid segment as on one straight line.

In other words

Let AB be a straight line, arbitrarily cut at point C

Then four times the rectangle formed by lines AB and BC plus the square of AC is equal to the square of AB added to BC

$AB = AC + CB$

$4 \cdot AB \cdot BC + AC^2 = (AB + BC)^2$

$AB = x + y$

$4(x + y)y + x^2 = (x + 2y)^2$

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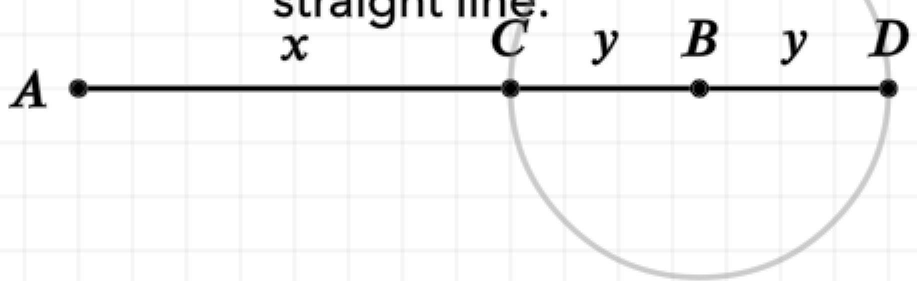

$AB = AC + CB\ ,\quad CB = BD$

In other words

Let AB be a straight line, arbitrarily cut at point C

$4 \cdot AB \cdot BC + AC^2 = (AB + BC)^2$

Construction:

Extend the line AB to point D such that CB is equal to BD

$AB = AC + CB$

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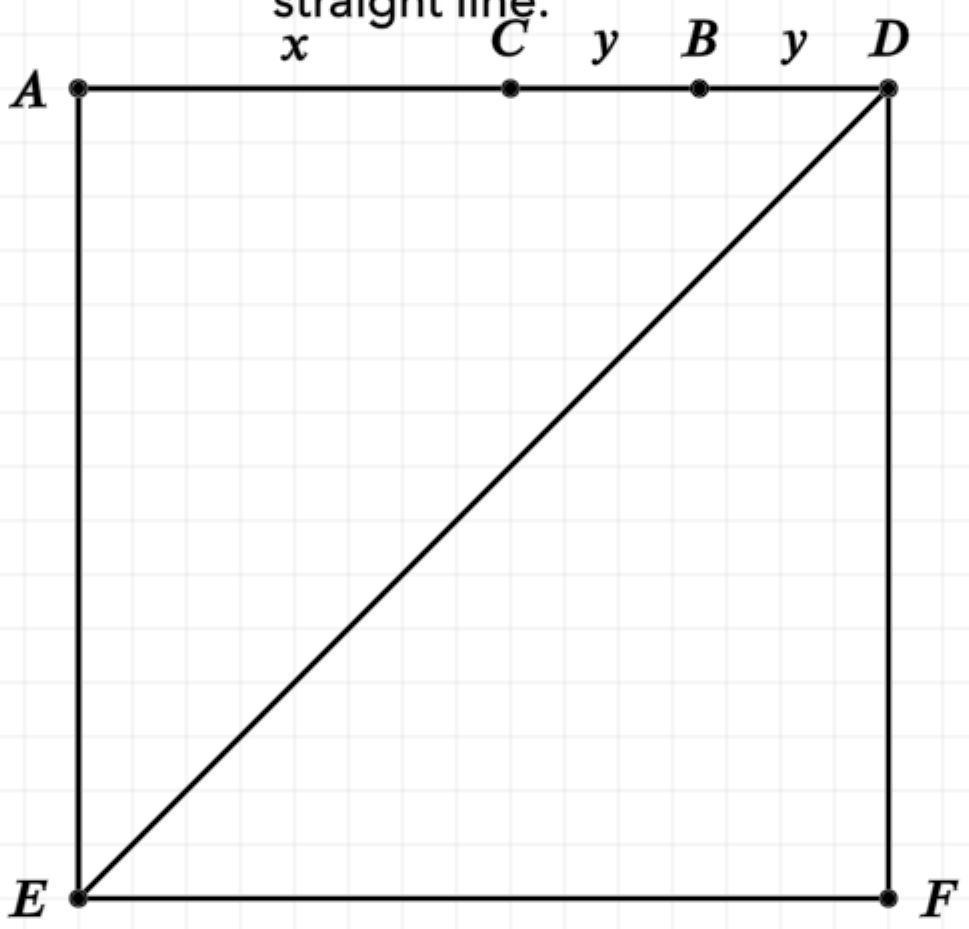

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Construction:

Extend the line AB to point D such that CB is equal to BD

Draw a square AEFD on the line AD (I.46), and draw the diagonal DE

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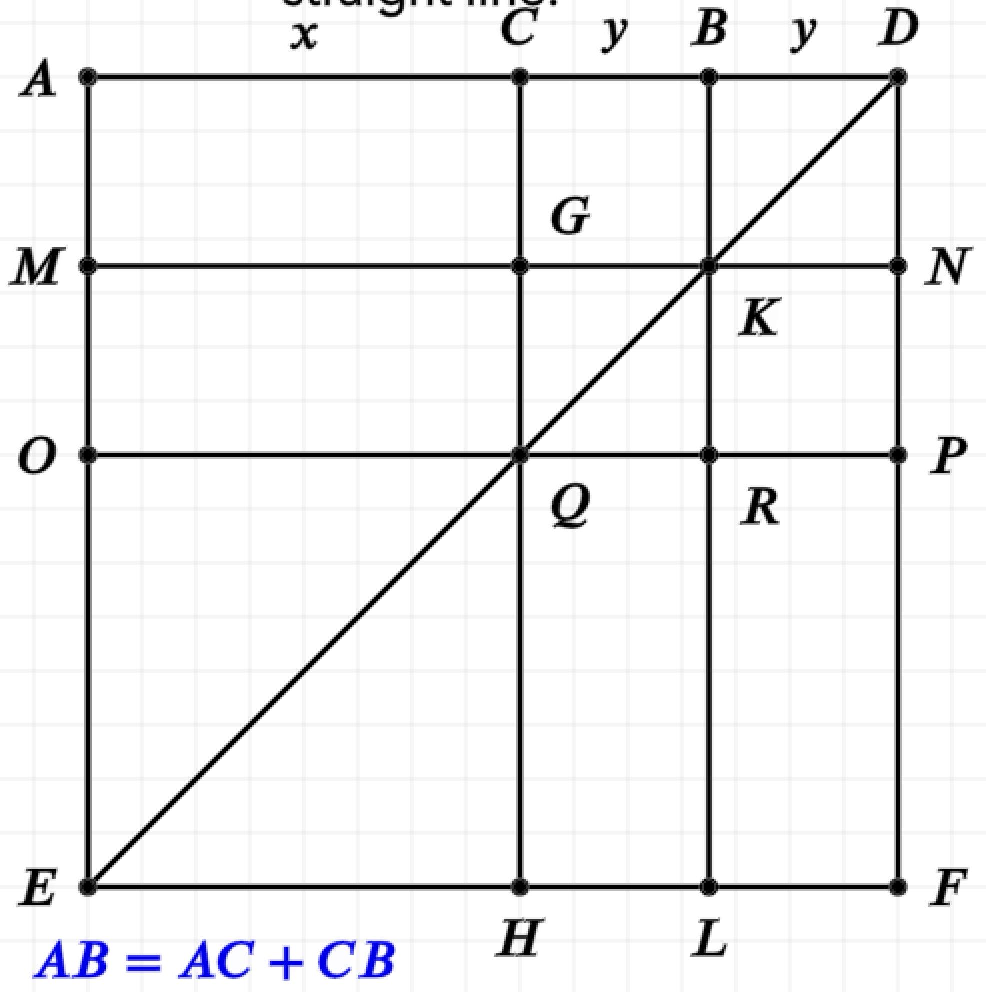

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Draw a square AEFD on the line AD (I.46), and draw the diagonal DE

Draw lines CH,BL parallel to AE (I.31)

Draw lines MN,OP parallel to AD (I.31)

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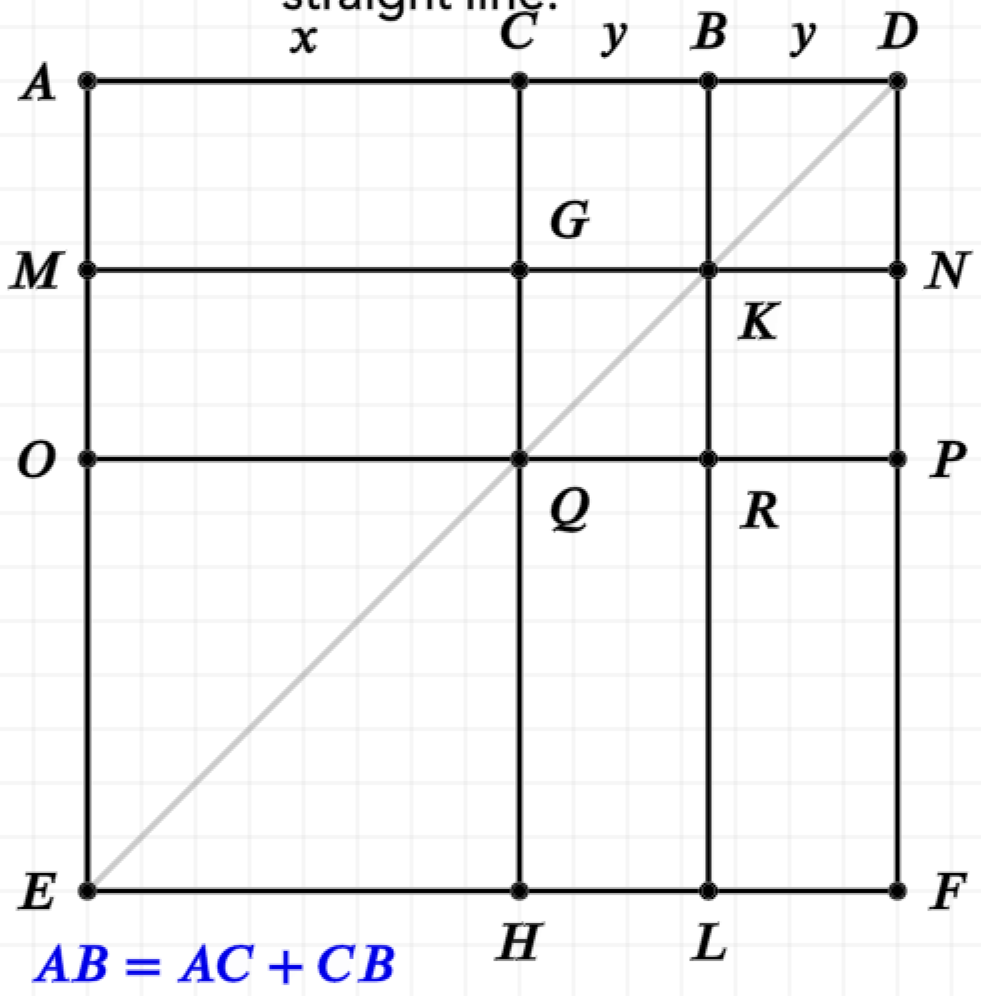

$AB = AC + CB$, $CB = BD$

$GK = KN$

$QR = RP$

In other words

Let AB be a straight line, arbitrarily cut at point C

$4 \cdot AB \cdot BC + AC^2 = (AB + BC)^2$

Proof:

Since CB is equal to BD, and CB is also equal to GK (I.34) and BD is equal to KN, then GK is equal to KN

Similarly, QR is equal to RP

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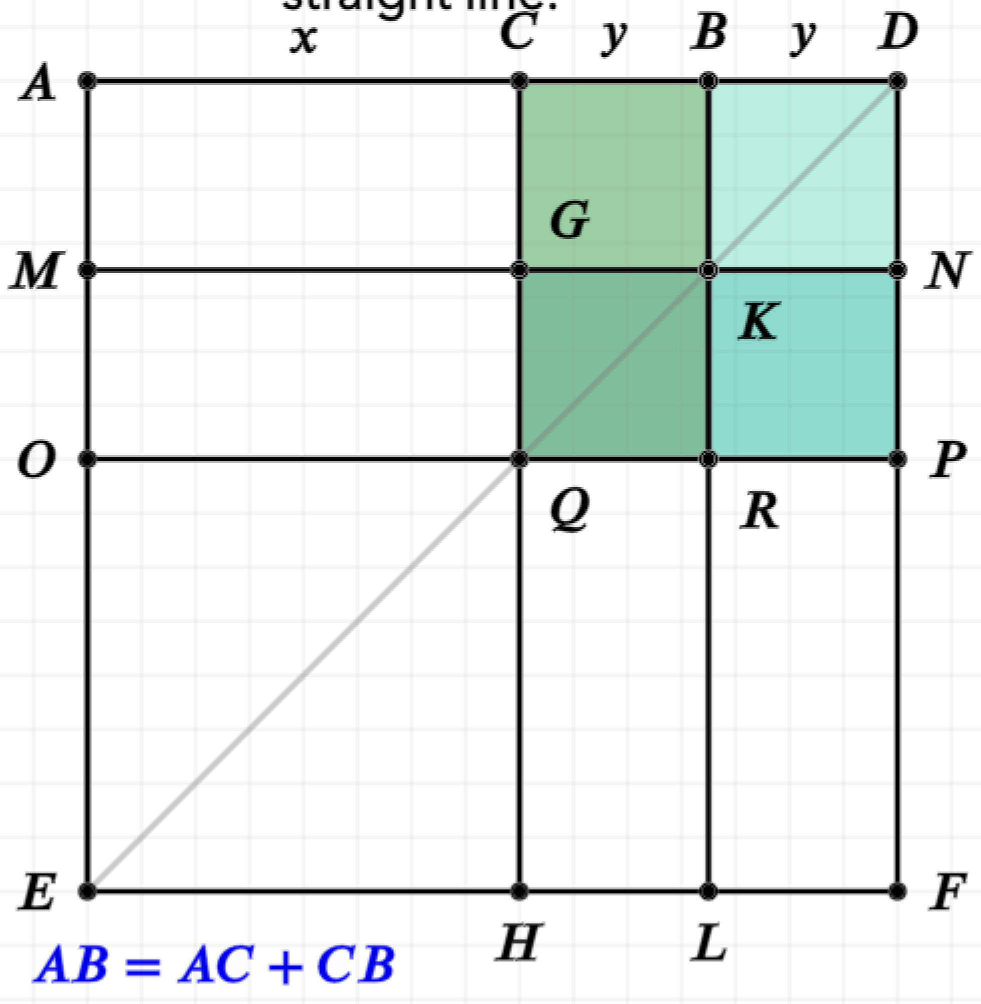

$AB = AC + CB$, $CB = BD$

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$CK = BN$

$GR = KP$

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Similarly, QR is equal to RP

Thus CK and BN are equal, as are GR and KP (I.36)

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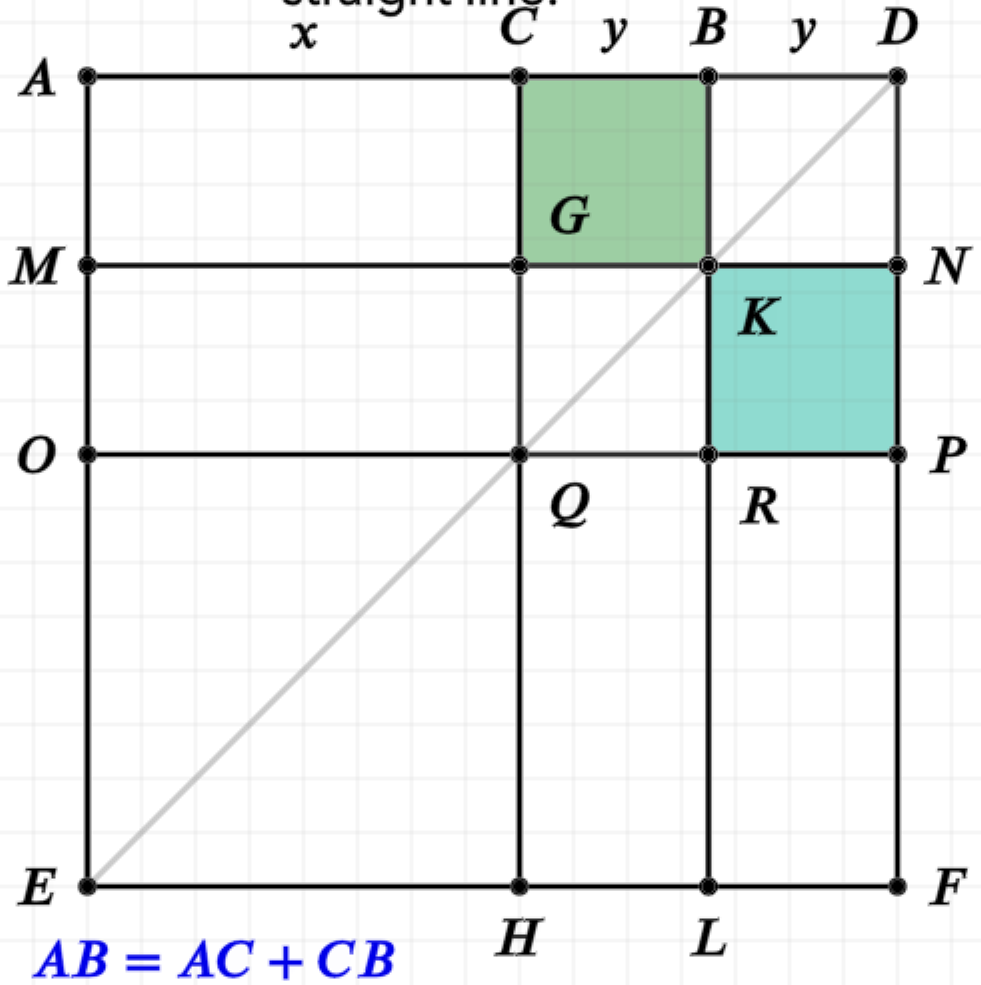

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CK and KP are equaL (I.43)

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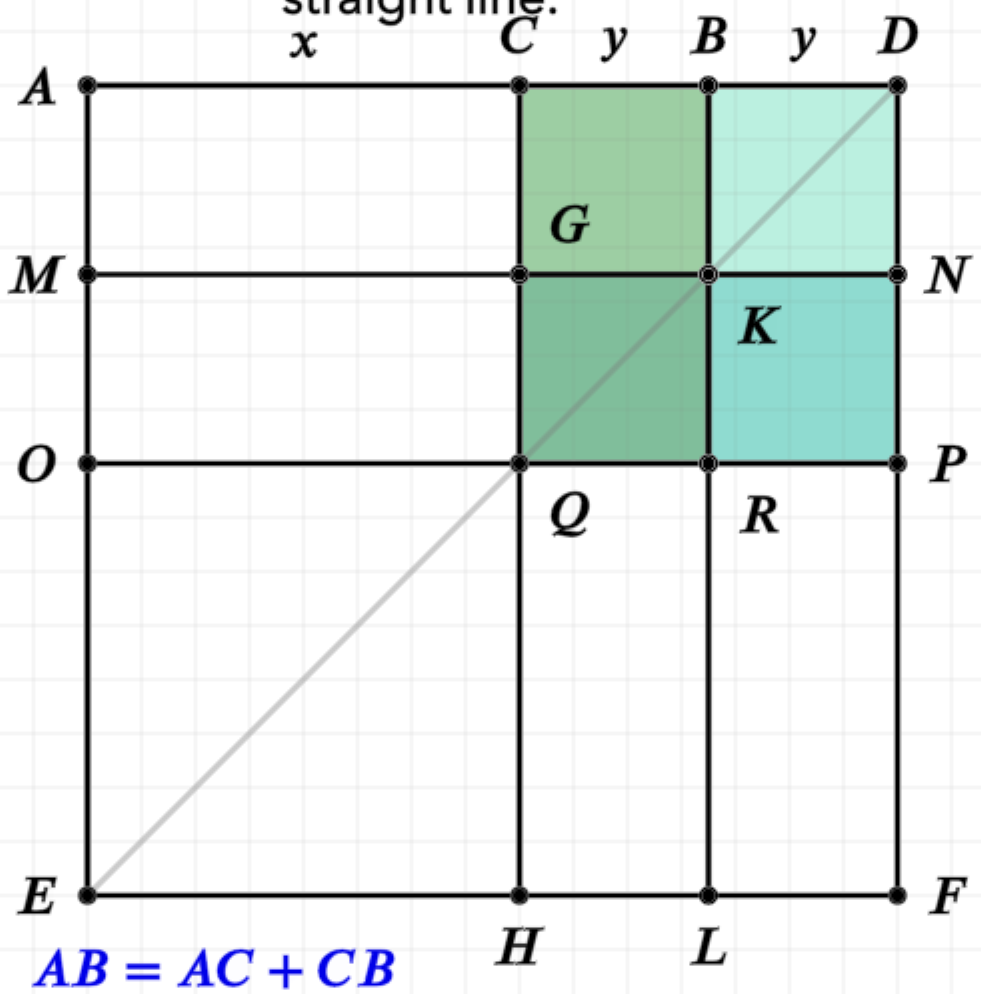

$AB = AC + CB$

$CK = BN = GR = KP$

$AB = AC + CB \; , \quad CB = BD$

$GK = KN$

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$GR = KP$

$CK = KP$

In other words

Let AB be a straight line, arbitrarily cut at point C

$$4 \cdot AB \cdot BC + AC^2 = (AB + BC)^2$$

Proof:

Therefore CK, BN, GR, KP are all equal, and the sum equals four CK

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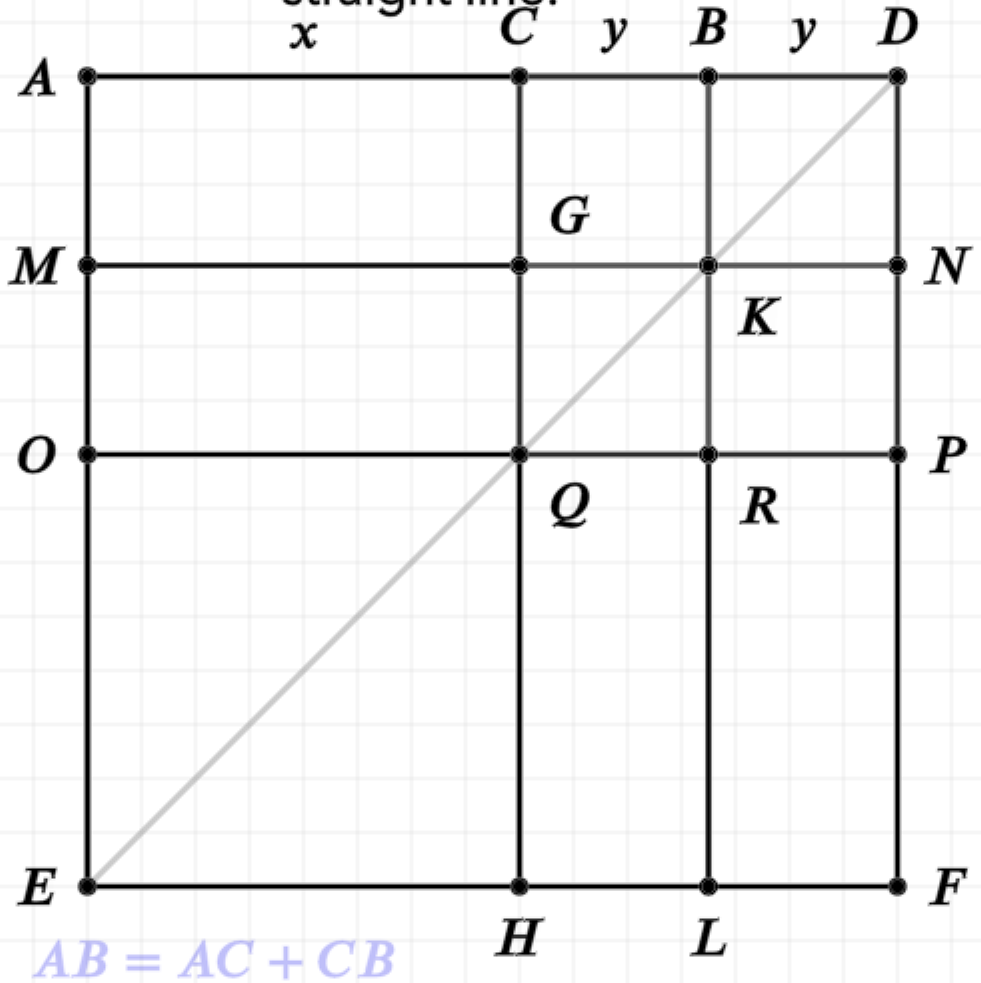

$AB = AC + CB$

$CK = BN = GR = KP$

$AB = AC + CB\ , \quad CB = BD$

$GK = KN$

$QR = RP$

$CK = BN$

$GR = KP$

$CK = KP$

$CG = GQ$

In other words

Let AB be a straight line, arbitrarily cut at point C

$4 \cdot AB \cdot BC + AC^2 = (AB + BC)^2$

Proof:

Therefore CK, BN, GR, KP are all equal, and the sum equals four CK

Again, since CB is equal to BD, and BD is also equal to BK which is equal to CG, and CB is equal to GK, which is equal to GQ, then CG is equal to GQ

Proposition 8 of Book 2

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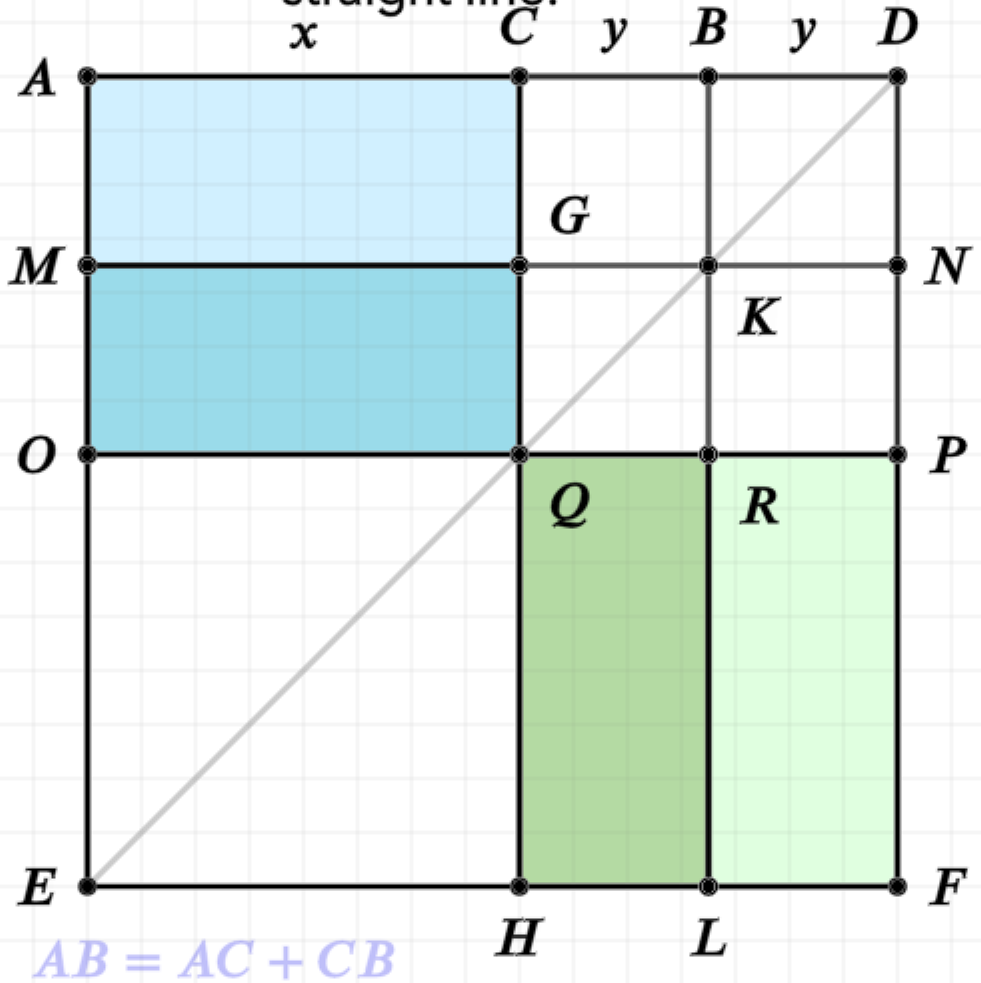

$AB = AC + CB$

$CK = BN = GR = KP$

$AB = AC + CB\ , \quad CB = BD$

$GK = KN$

$QR = RP$

$CK = BN$

$GR = KP$

$CK = KP$

$CG = GQ$

$AG = MQ$

$QL = RF$

In other words

Let AB be a straight line, arbitrarily cut at point C

$$4 \cdot AB \cdot BC + AC^2 = (AB + BC)^2$$

Proof:

Therefore CK, BN, GR, KP are all equal, and the sum equals four CK

Again, since CB is equal to BD, and BD is also equal to BK which is equal to CG, and CB is equal to GK, which is equal to GQ, then CG is equal to GQ

Thus AG and MQ are equal, as are QL and RF (I.36)

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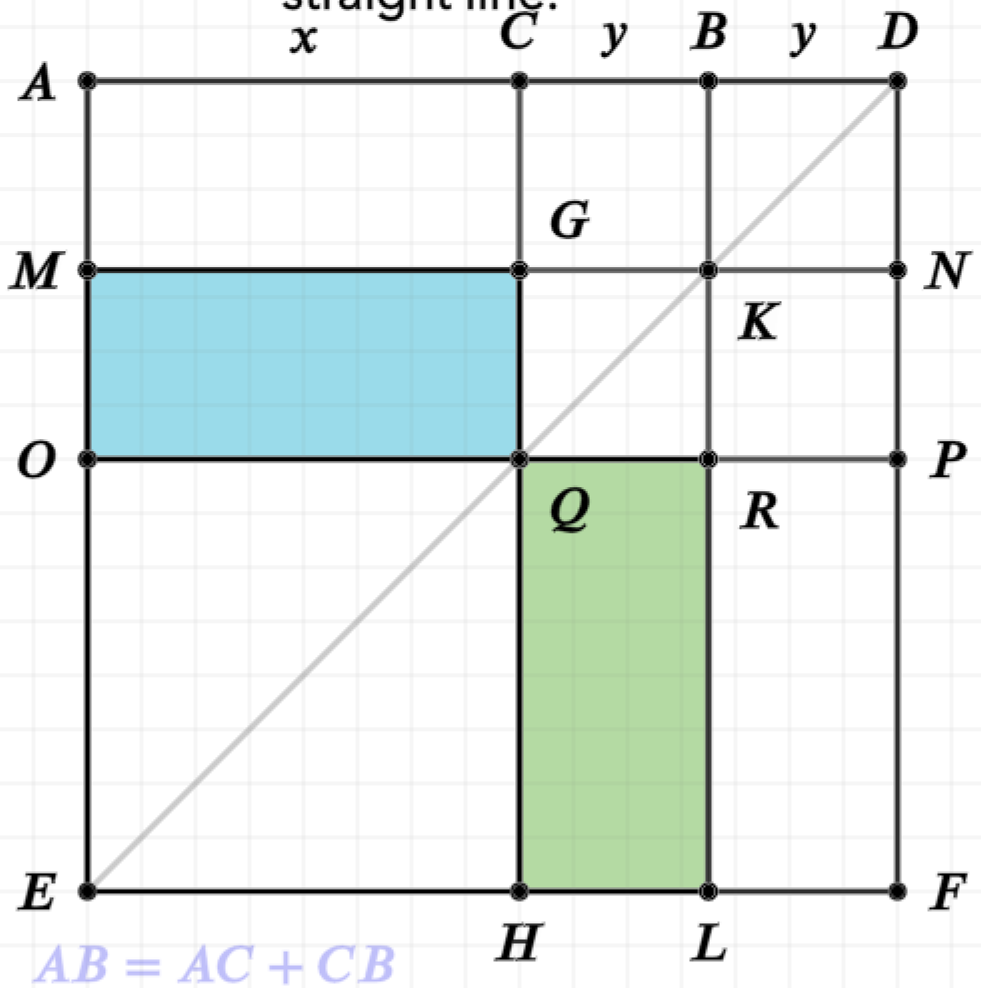
If a straight line be cut at random, four times the rectangle contained by the whole and one of the segments together with the square on the remaining segment is equal to the square described on the whole and aforesaid segment as on one straight line.



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In other words

Let AB be a straight line, arbitrarily cut at point C

$4 \cdot AB \cdot BC + AC^2 = (AB + BC)^2$

Proof:

Therefore CK, BN, GR, KP are all equal, and the sum equals four CK

Again, since CB is equal to BD, and BD is also equal to BK which is equal to CG, and CB is equal to GK, which is equal to GQ, then CG is equal to GQ

Thus AG and MQ are equal, as are QL and RF (I.36)

MQ and QL are equal (I.43)

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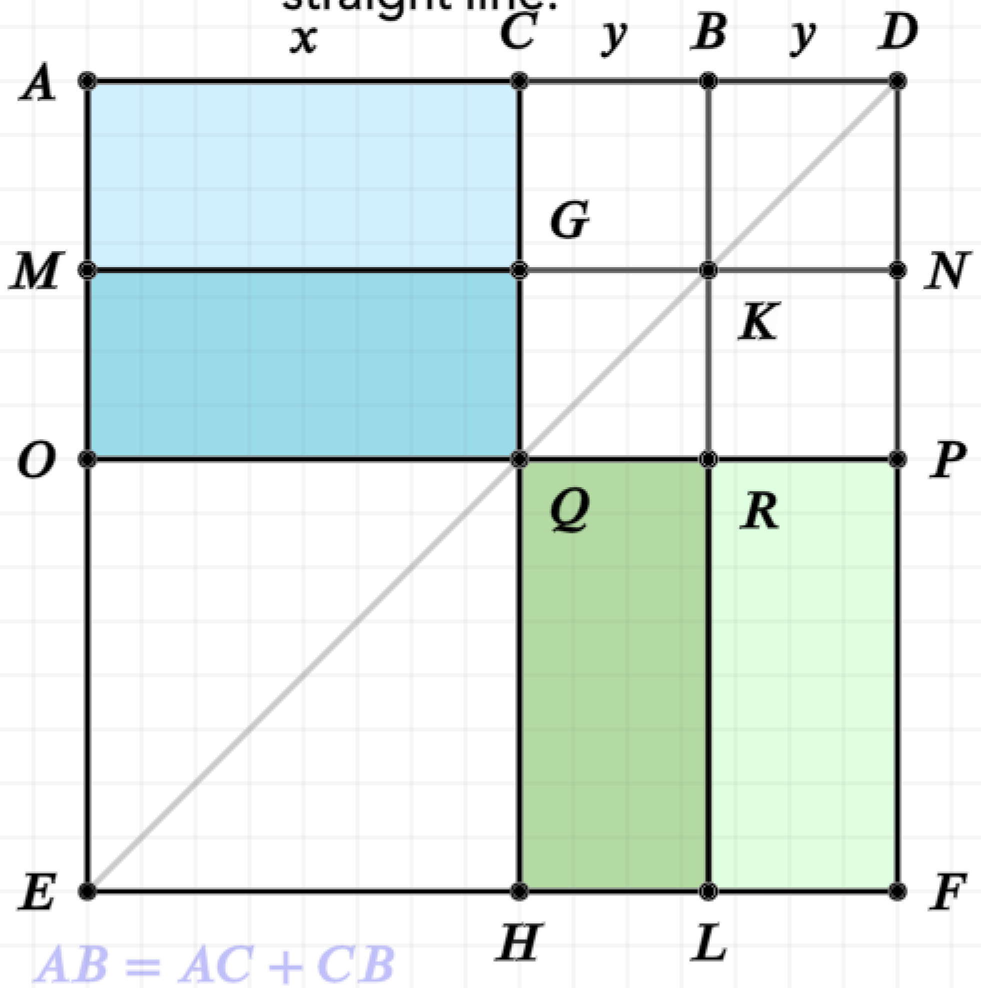
If a straight line be cut at random, four times the rectangle contained by the whole and one of the segments together with the square on the remaining segment is equal to the square described on the whole and aforesaid segment as on one straight line.



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$AB = AC + CB\ , \quad CB = BD$

$GK = KN$

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$CG = GQ$

$AG = MQ$

$QL = RF$

$MQ = QL$

In other words

Let AB be a straight line, arbitrarily cut at point C

$$4 \cdot AB \cdot BC + AC^2 = (AB + BC)^2$$

Proof:

Therefore CK, BN, GR, KP are all equal, and the sum equals four CK

Therefore MQ,QL,AG,RF are all equal, and the sum of the areas is four AG

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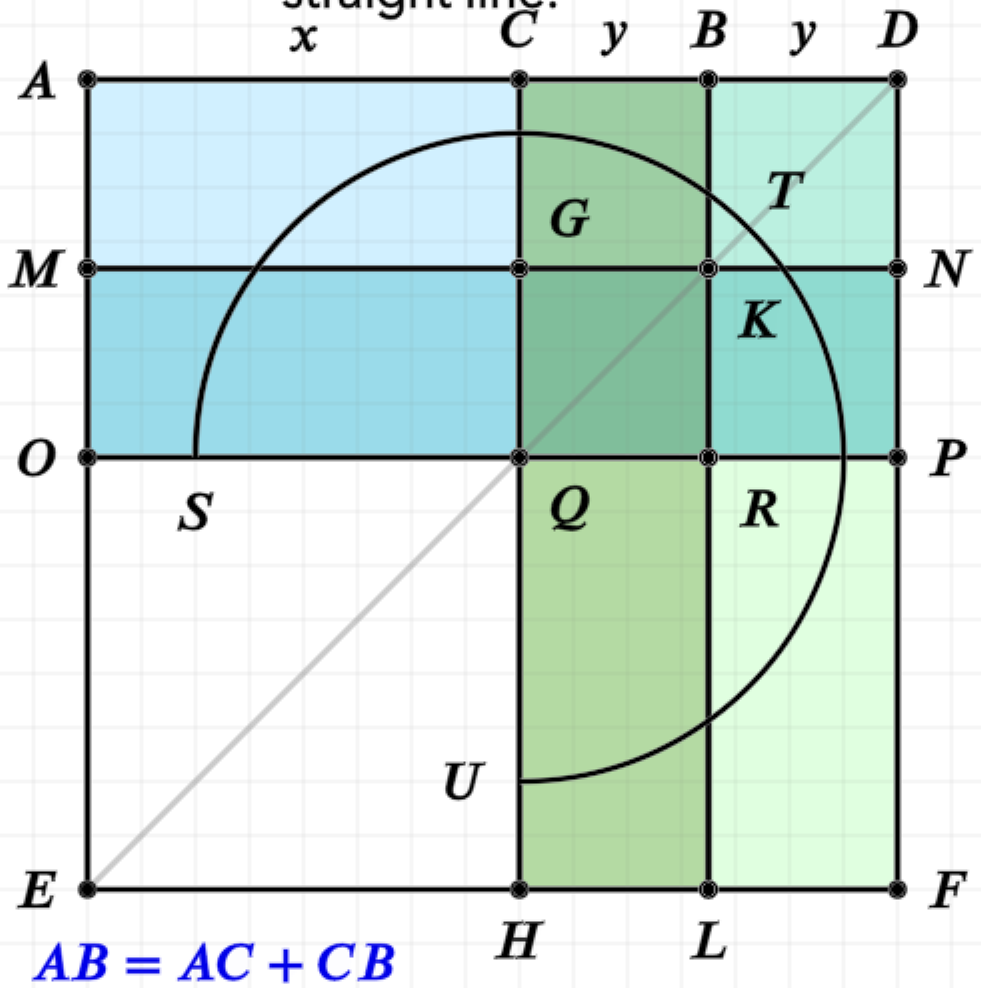

$AB = AC + CB$

$CK = BN = GR = KP$

$MQ = QL = AG = RF$

$STU = 4 \cdot (AG + CK) = 4 \cdot AK$

$AB = AC + CB\ , \quad CB = BD$

$GK = KN$

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Proof:

Therefore CK, BN, GR, KP are all equal, and the sum equals four CK

Therefore MQ,QL,AG,RF are all equal, and the sum of the areas is four AG

The gnomon STU is equal to the sum of all eight areas, which is also equal to four times AG plus CK

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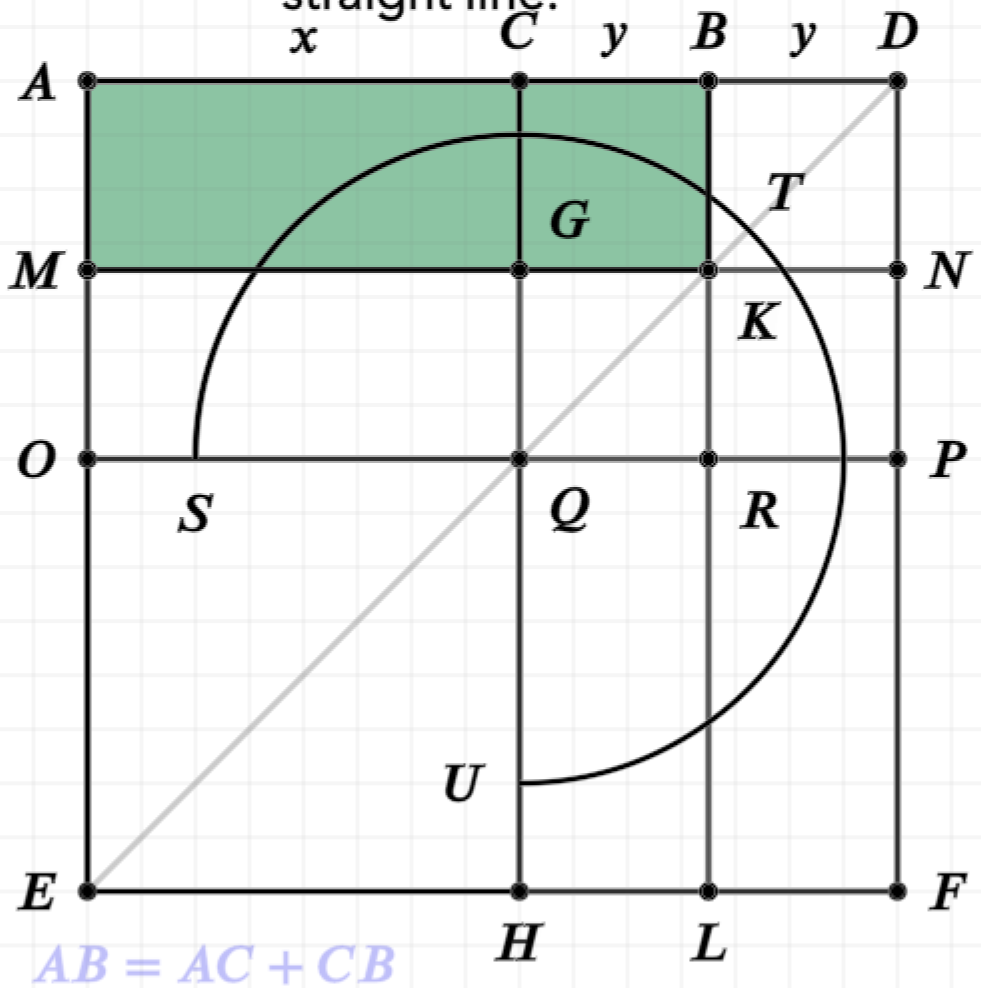

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In other words

Let AB be a straight line, arbitrarily cut at point C

$$4 \cdot AB \cdot BC + AC^2 = (AB + BC)^2$$

Proof:

Therefore CK, BN, GR, KP are all equal, and the sum equals four CK

Therefore MQ,QL,AG,RF are all equal, and the sum of the areas is four AG

The gnomon STU is equal to the sum of all eight areas, which is also equal to four times AG plus CK

AK is the rectangle formed by AB, BD (since BK equals BD), hence four AK is equal to four times AB, BD, which is also equal to the gnomon STU

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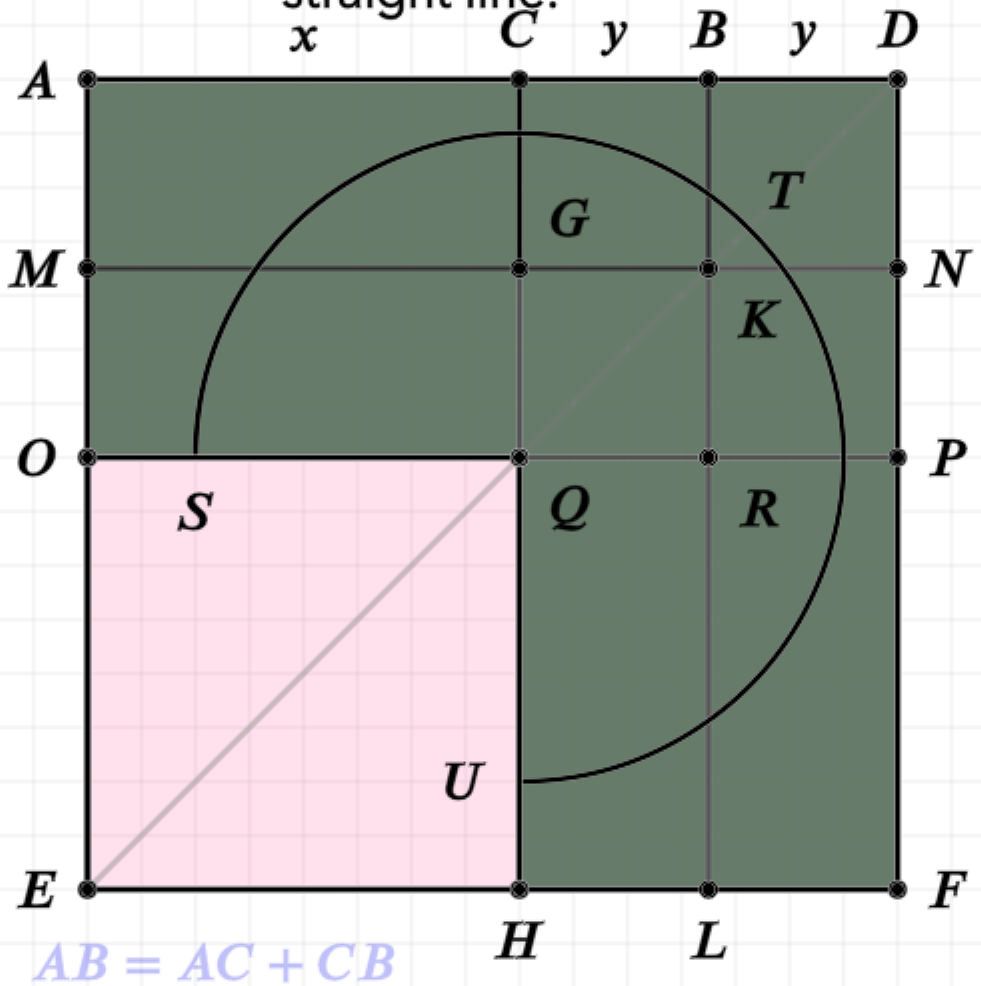

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$STU = 4 \cdot AK = 4 \cdot AB \cdot BD$

$STU + OH = AD^2 = 4 \cdot AB \cdot BD + AC^2$

$AB = AC + CB\ ,\quad CB = BD$

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Let AB be a straight line, arbitrarily cut at point C

$$4 \cdot AB \cdot BC + AC^2 = (AB + BC)^2$$

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Therefore MQ,QL,AG,RF are all equal, and the sum of the areas is four AG

The gnomon STU is equal to the sum of all eight areas, which is also equal to four times AG plus CK

AK is the rectangle formed by AB, BD (since BK equals BD), hence four AK is equal to four times AB, BD, which is also equal to the gnomon STU

Add the square of AC (which is also equal to OH) and we have four times the rectangle AB,BD plus the square of AC is equal to the gnomon plus OH, which is equal to the square of AD

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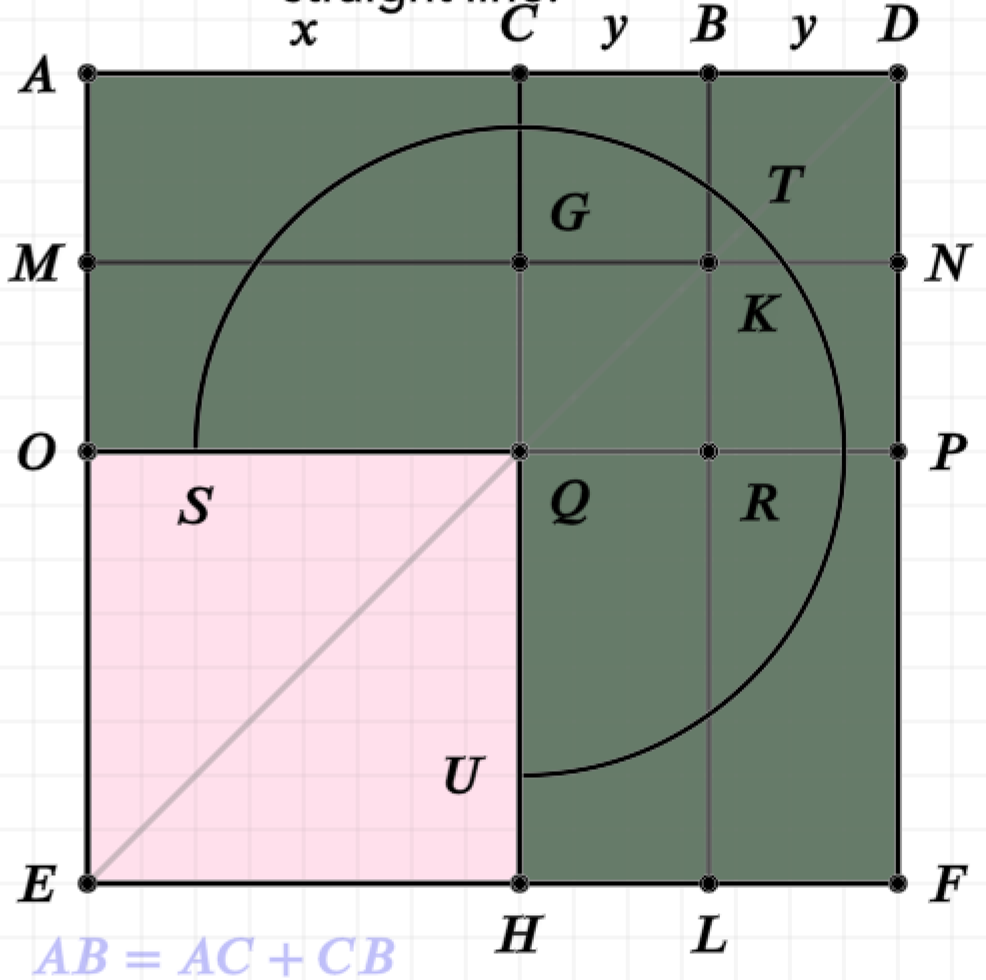

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Let AB be a straight line, arbitrarily cut at point C

$4 \cdot AB \cdot BC + AC^2 = (AB + BC)^2$

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Therefore CK, BN, GR, KP are all equal, and the sum equals four CK

Therefore MQ,QL,AG,RF are all equal, and the sum of the areas is four AG

The gnomon STU is equal to the sum of all eight areas, which is also equal to four times AG plus CK

AK is the rectangle formed by AB, BD (since BK equals BD), hence four AK is equal to four times AB, BD, which is also equal to the gnomon STU

Add the square of AC (which is also equal to OH) and we have four times the rectangle AB,BD plus the square of AC is equal to the gnomon plus OH, which is equal to the square of AD

And finally, since BD is equal to CB, and AD is equal to AB with BC added in a straight line, the square of AC added with quadruple the rectangle of AB and AC, is equal to the square of AB added to BC

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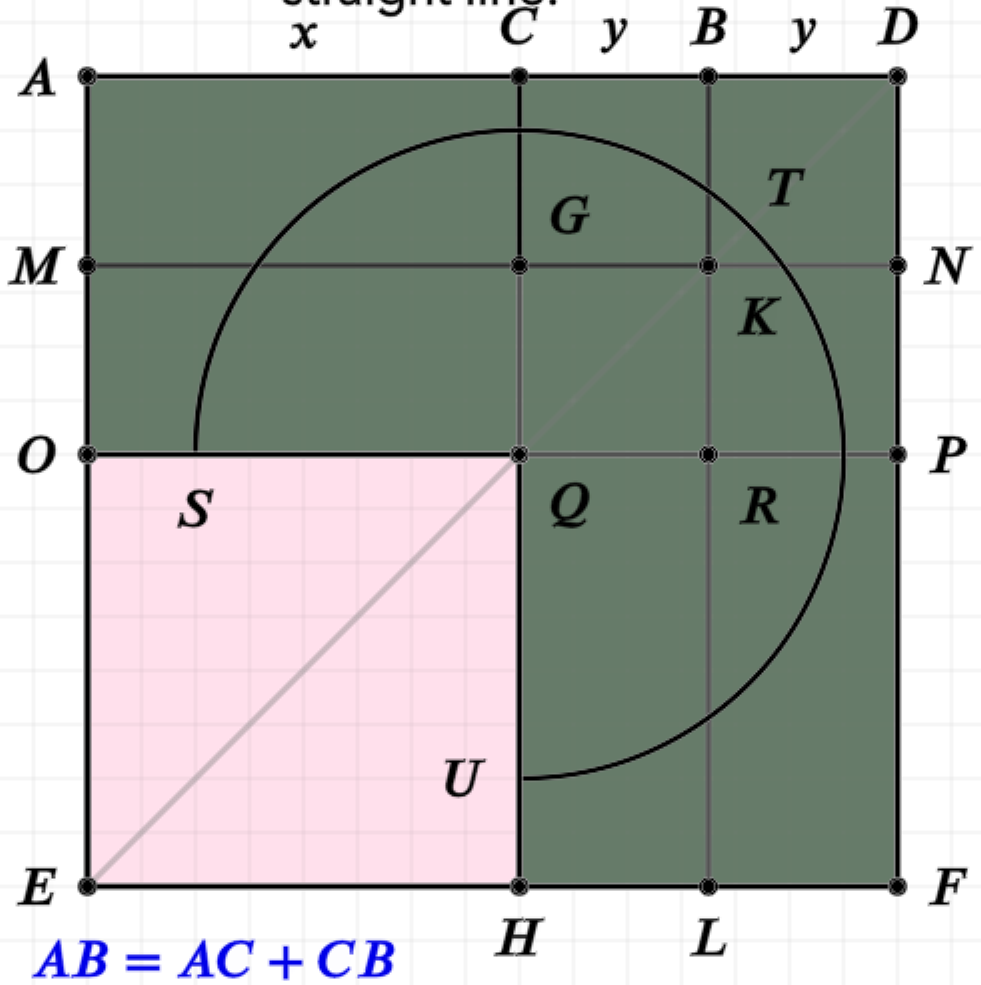
If a straight line be cut at random, four times the rectangle contained by the whole and one of the segments together with the square on the remaining segment is equal to the square described on the whole and aforesaid segment as on one straight line.



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The gnomon STU is equal to the sum of all eight areas, which is also equal to four times AG plus CK

AK is the rectangle formed by AB, BD (since BK equals BD), hence four AK is equal to four times AB, BD, which is also equal to the gnomon STU

Add the square of AC (which is also equal to OH) and we have four times the rectangle AB,BD plus the square of AC is equal to the gnomon plus OH, which is equal to the square of AD

And finally, since BD is equal to CB, and AD is equal to AB with BC added in a straight line, the square of AC added with quadruple the rectangle of AB and AC, is equal to the square of AB added to BC

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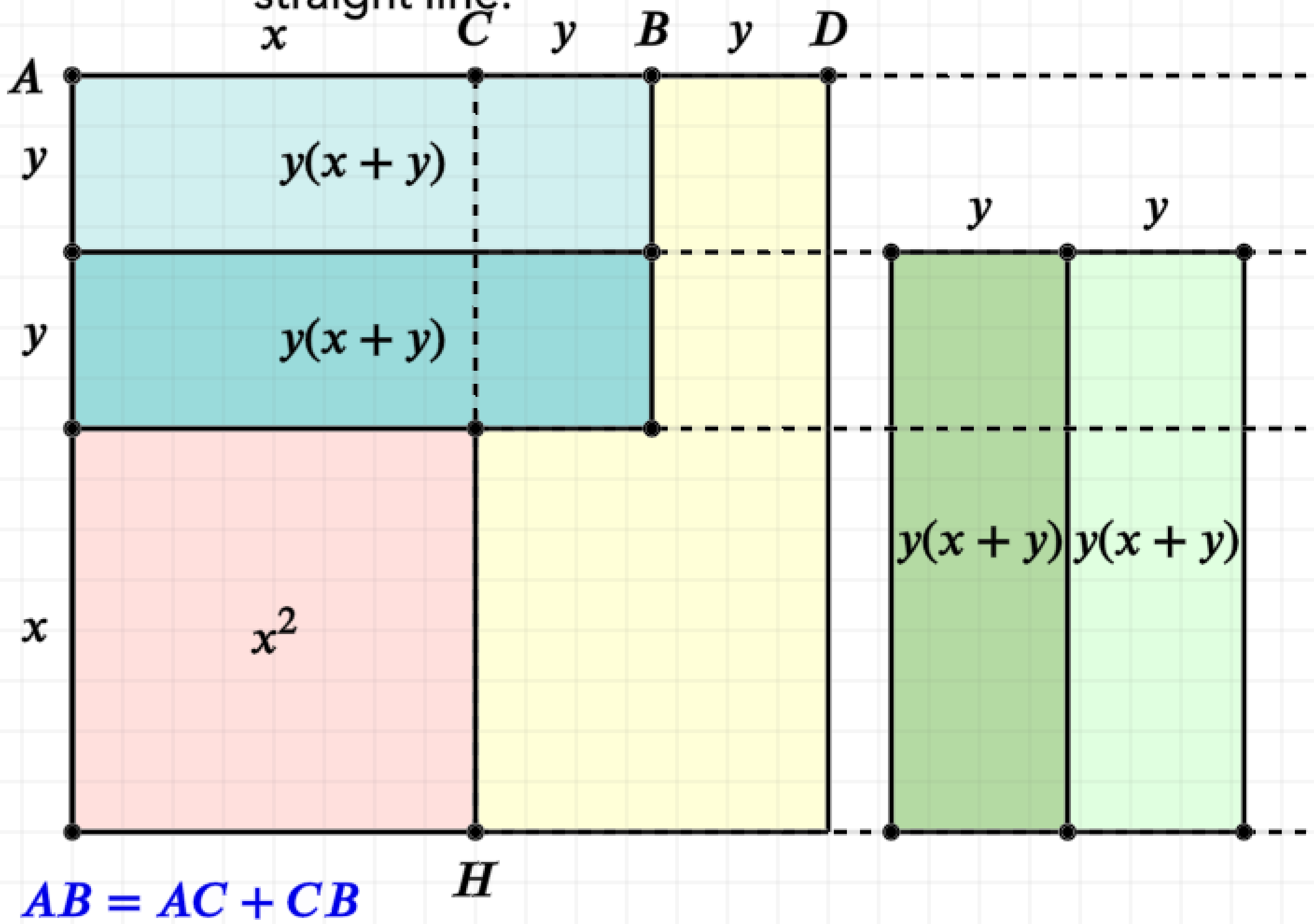

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$(AB + CB)^2 = 4 \cdot AB \cdot BC + AC^2$

$(x + 2y)^2 = 4y(x + y) + x^2$

In other words

Let AB be a straight line, arbitrarily cut at point C

$4 \cdot AB \cdot BC + AC^2 = (AB + BC)^2$

Proof:

Therefore CK, BN, GR, KP are all equal, and the sum equals four CK

Therefore MQ,QL,AG,RF are all equal, and the sum of the areas is four AG

The gnomon STU is equal to the sum of all eight areas, which is also equal to four times AG plus CK

AK is the rectangle formed by AB, BD (since BK equals BD), hence four AK is equal to four times AB, BD, which is also equal to the gnomon STU

Add the square of AC (which is also equal to OH) and we have four times the rectangle AB,BD plus the square of AC is equal to the gnomon plus OH, which is equal to the square of AD

And finally, since BD is equal to CB, and AD is equal to AB with BC added in a straight line, the square of AC added with quadruple the rectangle of AB and AC, is equal to the square of AB added to BC

Content:

Youtube Videos: *https://www.youtube.com/c/SandyBultena*

PDFs: https://github.com/sandy-bultena/Euclid/tree/python-manim/Propositions_PDFs

Additional Credits

Source code to create videos:

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